Worked Answer Key

1. **1:** $P = 2(8 + 3) =$ **22 cm**. **2:** $P = 4 \times 6 =$ **24 m**.
2. **1:** $P = 2(14 + 5) =$ **38 ft**. **2:** $36 \div 4 =$ **9 in** per side.
3. **1:** $2(12 + 9) =$ **42 m**. **2:** $7 + 8 + 10 =$ **25 in**.
4. **1:** $11 \times 4 =$ **44 cm²**. **2:** $15 \times 6 =$ **90 ft²**.
5. **1:** $9 \times 9 =$ **81 yd²**. **2:** $12 \times 12 = 144$, so the side is **12 in**.
6. **1:** $72 \div 9 =$ **8 cm**. **2:** $120 \div 8 =$ **15 ft**.
7. **1:** $18 \times 7 =$ **126 m²**. **2:** One poster is $2 \times 3 = 6$ ft²; $4 \times 6 =$ **24 ft²**.
8. **1:** Fencing is perimeter: $2(20 + 13) =$ **66 yd**. **2:** Carpet is area: $10 \times 12 =$ **120 ft²**.
9. **1:** $A = 16 \times 5 =$ **80 cm²**; $P = 2(16 + 5) =$ **42 cm**. **2:** $A = 7 \times 7 =$ **49 m²**; $P = 4 \times 7 =$ **28 m**.
10. **1:** $10 \times 4 + 6 \times 3 = 40 + 18 =$ **58 cm²**. **2:** $8 \times 5 + 4 \times 3 = 40 + 12 =$ **52 ft²**.
11. **1:** $5 \times 3 \times 2 =$ **30 unit cubes** fill the prism; its volume is 30 cubic units. **2:** $7 \times 2 \times 4 =$ **56 cubic units**.
12. **1:** $9 \times 4 \times 3 =$ **108 cm³**. **2:** $12 \times 5 \times 2 =$ **120 in³**.
13. **1:** Base area $8 \times 4 = 32$ m²; $96 \div 32 =$ **3 m**. **2:** Width × height $= 6 \times 5 = 30$ ft²; length $= 180 \div 30 =$ **6 ft**.
14. **1:** $10 \times 4 \times 5 =$ **200 dm³**. **2:** $8 \times 6 \times 5 =$ **240 in³**.
15. **1:** One layer has $6 \times 4 = 24$ unit cubes; $24 \times 3 = 72$ unit cubes, so the volume is **72 cubic units**. **2:** $48 \times 5 =$ **240 cubes**.
16. **1:** $3 \times 2 =$ **6 ft²**. **2:** 200 cm = 2 m; $2 \times 3 =$ **6 m²**.
17. **1:** Garden perimeter $2(9 + 6) = 30$ m. Subtract the gate's 2 m width: $30 - 2 =$ **28 m**. The gate's height does not affect the fence length. **2:** $4 \times 4 \times 4 =$ **64 cm³**. Each unit cube measures 1 cm × 1 cm × 1 cm, so its volume is 1 cubic centimeter.
18. **1:** Playground $25 \times 14 = 350$; sandbox $3 \times 4 = 12$; $350 - 12 =$ **338 m²**. **2:** $12 \times 10 \times 6 =$ **720 cubes**.
19. **1:** Example: 6 cm × 8 cm has area 48 cm². Its perimeter is $2(6 + 8) =$ **28 cm**; the 12 × 4 rectangle has $P = 2(12 + 4) =$ **32 cm**, so the new perimeter is 4 cm smaller. Other whole-number pairs are valid if their product is 48 and their perimeters are compared correctly. **2:** Original $V = 3 \times 4 \times 5 = 60$ ft³. Doubling one dimension doubles the number of unit cubes that fit inside, so the new volume is $2 \times 60 =$ **120 ft³**.
20. **1:** $P = 4 \times 11 =$ **44 m**; $A = 11 \times 11 =$ **121 m²**. **2:** $V = 15 \times 4 \times 3 =$ **180 cm³**; top area $= 15 \times 4 =$ **60 cm²**.